Efficient direct localization of overlapping wideband sources with unequal powers: multi-candidate joint refinement and waveform-eliminated updates

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ABSTRACT

Spectral overlap and unequal received powers complicate the extraction and association of source-specific delays in two-step localization. Direct position determination instead fits the mixed station observations jointly, but a tractable solver must choose candidate regions and refine positions coupled to unknown waveforms. This paper addresses these two operations for synchronized single-antenna stations with known relative amplitudes and baseband delays. Multi-candidate joint-refinement alternating projection (MCJ-AP) retains separated conditional peaks and selects a branch only after jointly refining both positions. Waveform-eliminated block Newtonized orthogonal matching pursuit (VP-BNOMP) incorporates current-source waveform elimination into the local position curvature. On 2,400 observations from 20 source configurations, five-candidate MCJ-AP raises two-source success within 20 m from 95% to 99%; the additional recoveries occur in one configuration and mostly require the fifth-ranked candidate. Joint refinement reduces mean complete runtime by 31.36% with the same candidates and projected-gradient stopping tolerance. A separate 240-observation pair isolates an 83.77% mean runtime reduction from reduced curvature under a common waveform-eliminated acceptance objective. Tests at new source positions show when extra AP candidates add cost without recovery and when VP retains localization failures. The results quantify candidate retention's recovery–cost tradeoff and the local computational benefit of waveform-eliminated curvature.

1. Introduction

Emitter localization often estimates time differences of arrival before solving for position. Generalized correlation and hyperbolic localization provide established tools for this two-step approach [3, 9]. When unknown emitters occupy the same frequency band, each station receives a mixture with different propagation responses. Extracting and associating individual delays then becomes difficult, particularly for a weak emitter in the presence of a stronger one. Direct position determination (DPD) incorporates propagation into the observation model and estimates positions from the station data jointly [5, 15].

This paper considers two independent unknown wideband sources observed by four synchronized single-antenna stations. The positions, transmitted waveforms and powers are unknown; station coordinates and relative propagation responses are available. A single-source spectrum can assign several large peaks to the stronger source, whereas an exhaustive search over two two-dimensional positions requires many combinations. Sequential discovery and continuous refinement reduce this search. Their remaining difficulties occur at different stages: discarding a candidate

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can exclude an effective refinement branch, while coupling between positions and unknown waveforms can make local updates expensive.

Wideband maximum likelihood and alternating projection (AP) reduce joint search dimensionality by updating one position conditional on the others [5]. MUSIC-type DPD uses orthogonality to an estimated noise subspace [15]. MVDR uses the inverse covariance to suppress interfering components [2], and DPD-HR applies this principle to direct localization [14]. These methods process mixtures differently, and their relative performance also depends on covariance estimation and search. We therefore compare concrete implementations using the same data and propagation information.

Newtonized orthogonal matching pursuit (NOMP) combines greedy discovery, Newton refinement and cyclic updates [11]. The two-dimensional multi-frequency block NOMP of Yang et al. groups the unknown frequency coefficients at a common position and re-estimates them after position updates [16]. It supplies a close reference for the local update studied here. QNOMP eliminates path gains and jointly refines delays and angles from pilot-based channel observations [17]. In our model the unknown coefficients are the per-frequency amplitudes of independent source waveforms, and the nonlinear parameters are source positions.

Candidate selection also has relevant precedents. Oh and Un form a sequential MUSIC initialization and then apply AP updates [13]. LAOLS compares discrete atom candidates through residuals after tentative support completion [4]. MCJ-AP uses conditional projection scores to retain separated spatial peaks, then continuously refines both source positions for each retained branch. The conditional score already eliminates both waveforms, but fixes the first position; joint fitting also allows that position to move. Thus a lower-ranked conditional peak may yield a better jointly fitted position pair. MCJ-AP delays branch selection until this fit, without repeating the full spatial scan for each branch.

The treatment of unknown waveforms affects refinement cost as well. Variable projection eliminates linear coefficients before optimizing nonlinear parameters [7]. We incorporate the resulting frequency-separable curvature into local block-NOMP position updates and compare it with fixed-coefficient curvature under the same acceptance objective. The two methods address candidate selection and local update cost within the same wideband localization model; they are separate solvers. The contributions are:

1. MCJ-AP postpones branch selection until after continuous joint fitting. Saved-branch analysis identifies the recoveries supplied by lower-ranked conditional peaks, and a matched-candidate comparison separates this coverage benefit from the computational savings of joint over alternating refinement.
2. VP-BNOMP uses waveform-eliminated position curvature while retaining residual discovery and cyclic block updates. Its frequency-wise small-matrix construction and local curvature analysis explain the update, and a common-acceptance comparison isolates its effect on refinement cost from complete localization reliability.

Experiments compare eight method families on a common 240-observation subset, examine the principal methods on 2,400 observations, and test frozen configurations on 480 observations at new source positions. The receiver coordinates remain fixed. These experiments distinguish candidate recovery from local computational savings.

2. Observation model and localization objective

2.1. Synchronized wideband observations

Let $\boldsymbol{r}_l \in \mathbb{R}^2$ denote station l, $l = 1, \dots, L$, and let $\boldsymbol{p}_k \in \Omega \subset \mathbb{R}^2$ denote source k, $k = 1, \dots, K$. We assume static direct-path propagation, synchronized station timing and frequency, calibrated relative receiver responses, and matched antenna polarization with equal directional responses. The experiments use $L = 4$ and known $K = 2$. Define

$$d_{lk} = \|\boldsymbol{p}_k - \boldsymbol{r}_l\|_2, \qquad \tau_{lk} = d_{lk}/c_0, \qquad \beta_{lk} = d_{lk}^{-1}, \tag{1}$$

where c_0 is propagation speed. For far-field, matched isotropic antennas, free-space attenuation gives

$$P_\mathrm{r}/P_\mathrm{t} = (\lambda/4\pi d)^2, \qquad a_\mathrm{amp}(d) = \sqrt{P_\mathrm{r}/P_\mathrm{t}} = \lambda/(4\pi d). \tag{2}$$

Here P_r and P_t are received and transmitted powers, and λ is wavelength [8]. With fixed λ and antenna responses, the common factor $\lambda/(4\pi)$ is absorbed into the unknown waveform, leaving relative amplitudes proportional to $1/d$.

The propagation phase is specified by an ideal baseband pure-delay model. Receiver calibration removes station responses; it does not establish this propagation assumption. Both generator and estimators use the same baseband model, without an RF carrier phase or random link gains. With sampling rate f_s and N samples,

$$x_l[n] = \sum_{k=1}^{K} \beta_{lk} s_k(n/f_s - \tau_{lk}) + w_l[n], \quad 0 \le n < N, \tag{3}$$

where s_k includes unknown transmitted power and the common amplitude factor. The source waveforms are independent. Noise samples are independent across stations and time, independent of the sources, and distributed as $\mathcal{CN}(0, \sigma_t^2)$.

The discrete Fourier transform (DFT) is

$$X_l(f_m) = N^{-1} \sum_{n=0}^{N-1} x_l[n] e^{-\mathrm{j}2\pi mn/N}, \qquad f_m = m f_s/N, \tag{4}$$

with signed frequency index m. Retaining the common band $\mathcal{F}$, with $F = |\mathcal{F}|$ bins, gives

$$\boldsymbol{x}_f = \boldsymbol{A}_f(\boldsymbol{P})\boldsymbol{s}_f + \boldsymbol{n}_f, \tag{5}$$

$$[\boldsymbol{a}_f(\boldsymbol{p})]_l = d_l(\boldsymbol{p})^{-1} \exp[-\mathrm{j}2\pi f d_l(\boldsymbol{p})/c_0], \tag{6}$$

$$\boldsymbol{A}_f(\boldsymbol{P}) = [\boldsymbol{a}_f(\boldsymbol{p}_1), \dots, \boldsymbol{a}_f(\boldsymbol{p}_K)].$$

Here $\boldsymbol{x}_f, \boldsymbol{n}_f \in \mathbb{C}^L$, $\boldsymbol{s}_f \in \mathbb{C}^K$, and $\boldsymbol{P} = [\boldsymbol{p}_1, \dots, \boldsymbol{p}_K] \in \mathbb{R}^{2\times K}$. Frequency f is the signed baseband frequency and $\mathrm{j}^2 = -1$. The common-window approximation neglects station-dependent edge differences [5], examined in Section 4.7. Band selection leaves the noise variance $\sigma_f^2 = \sigma_t^2/N$ and DFT normalization unchanged.

Unit-norm responses $\boldsymbol{q}_f = \boldsymbol{a}_f/\|\boldsymbol{a}_f\|_2$ may be used computationally. A common scaling of one response column is absorbed by its free coefficient and preserves relative station amplitudes. The implementation protects distances below 1 m; all simulated sources lie far outside that region. Superscripts H, T and $\dagger$ denote conjugate transpose, transpose and pseudoinverse, respectively.

Let P_{lk} be the time-averaged power of isolated source k at station l, and let B be nominal signal bandwidth. We define

$$\mathrm{SNR}_{k,\mathrm{in}} = 10\log_{10}\frac{L^{-1}\sum_l P_{lk}}{\sigma_t^2 B/f_s}, \qquad \Delta = 10\log_{10}\frac{\sum_l P_{l1}}{\sum_l P_{l2}}. \tag{7}$$

Source 2 and the noise are fixed when scaling source 1. Thus changing Δ leaves the weak-source SNR unchanged. For $\Delta > 0$ the sources are called strong and weak; at $\Delta = 0$ their identities remain A and B. Local power differences at individual stations need not equal Δ.

2.2. Eliminating the unknown waveforms

Treat the frequency coefficients as deterministic unknowns. Under the independent equal-variance complex Gaussian noise model, maximum likelihood is equivalent to least squares. For observation energy $E_x = \sum_f \|\boldsymbol{x}_f\|^2 > 0$, define

$$\mathcal{L}(\boldsymbol{P}, \{\boldsymbol{s}_f\}) = E_x^{-1}\sum_f \|\boldsymbol{x}_f - \boldsymbol{A}_f(\boldsymbol{P})\boldsymbol{s}_f\|^2, \tag{8}$$

$$\widehat{\boldsymbol{s}}_f(\boldsymbol{P}) = \boldsymbol{A}_f(\boldsymbol{P})^{\dagger}\boldsymbol{x}_f, \tag{9}$$

$$\boldsymbol{J}(\boldsymbol{P}) = E_x^{-1}\sum_f \|\boldsymbol{\Pi}^{\perp}_{\boldsymbol{A}_f}\boldsymbol{x}_f\|^2, \quad \boldsymbol{\Pi}^{\perp}_{\boldsymbol{A}_f} = \boldsymbol{I} - \boldsymbol{A}_f\boldsymbol{A}_f^{\dagger}. \tag{10}$$

Substitution of the least-squares waveforms yields the position-only objective J [7]. It measures the fraction of observation energy unexplained by the position set. Two-source localization then involves four real position variables, with coefficients fitted at every trial. The single-source score is

$$Q_0(\boldsymbol{p}) = E_x^{-1} \sum_f \frac{|\boldsymbol{a}_f(\boldsymbol{p})^H \boldsymbol{x}_f|^2}{\|\boldsymbol{a}_f(\boldsymbol{p})\|^2}. \tag{11}$$

For one source, $J = 1 - Q_0$. Taking two peaks from Q_0 does not jointly fit two sources; the subsequent algorithms use a multiple-source objective to account for their interaction.

3. Candidate retention and waveform-eliminated refinement

3.1. Conditional search and MCJ-AP

Given responses $\boldsymbol{B}_f$ at already selected positions, project both the data and trial response:

$$\boldsymbol{r}_f = \boldsymbol{\Pi}^{\perp}_{\boldsymbol{B}_f} \boldsymbol{x}_f, \qquad \boldsymbol{u}_f(\boldsymbol{p}) = \boldsymbol{\Pi}^{\perp}_{\boldsymbol{B}_f} \boldsymbol{a}_f(\boldsymbol{p}). \tag{12}$$

The conditional AP score is

$$Q_{\mathrm{AP}}(\boldsymbol{p} \mid \boldsymbol{B}) = E_x^{-1} \sum_f \frac{|\boldsymbol{u}_f(\boldsymbol{p})^H \boldsymbol{r}_f|^2}{\|\boldsymbol{u}_f(\boldsymbol{p})\|^2}. \tag{13}$$

For $\boldsymbol{u}_f \neq 0$, orthogonal projection gives

$$\boldsymbol{\Pi}^{\perp}_{[\boldsymbol{B}_f, \boldsymbol{a}_f]} = \boldsymbol{\Pi}^{\perp}_{\boldsymbol{B}_f} - \frac{\boldsymbol{u}_f \boldsymbol{u}_f^H}{\|\boldsymbol{u}_f\|^2}, \qquad J_{\mathrm{B}} - J([\boldsymbol{B}, \boldsymbol{a}(\boldsymbol{p})]) = Q_{\mathrm{AP}}, \tag{14}$$

where $J_{\mathrm{B}} = E_x^{-1} \sum_f \|\boldsymbol{r}_f\|^2$. With existing positions fixed, the highest conditional score gives the greatest joint residual reduction [5]. Computation uses normalized columns, sets a bin's score to zero when its projected norm squared is at most 10^{-20}, and uses an SVD pseudoinverse with relative cutoff 10^{-10}. The projection identity applies away from these numerical rank cutoffs.

MCJ-AP first maximizes Q_0 on the common grid $\mathcal{G}$, then continuously refines the first source to $\boldsymbol{p}_1^{(0)}$. With that position fixed, it scans Q_{AP} and retains M local maxima $\boldsymbol{c}_m$ separated by at least Δ_c. Each supplies a joint initial position pair:

$$\widehat{\boldsymbol{P}}^{(m)} = \mathrm{LocalMin}_{\boldsymbol{P} \in \Omega^2} J(\boldsymbol{P}; [\boldsymbol{p}_1^{(0)}, \boldsymbol{c}_m]), \qquad m^* = \arg\min_m J(\widehat{\boldsymbol{P}}^{(m)}). \tag{15}$$

The output is $\widehat{\boldsymbol{P}}^{(m^*)}$. The conditional score ranks candidates with the first position fixed; joint refinement moves both positions and re-estimates both waveforms. MCJ-AP therefore retains a candidate set at the grid stage and makes the final branch choice using the refined J. A larger nested set cannot increase the selected residual when existing branch solutions remain unchanged. This property concerns the fitted objective; localization errors are evaluated separately.

Joint refinement uses bounded L-BFGS-B [1]. Write $\boldsymbol{\theta} = \mathrm{vec}(\boldsymbol{P})$ and $\boldsymbol{e}_f = \boldsymbol{x}_f - \boldsymbol{A}_f \widehat{\boldsymbol{s}}_f$. When $\boldsymbol{A}_f$ is locally full column rank, $\boldsymbol{A}_f^H \boldsymbol{e}_f = 0$ eliminates the coefficient derivative from

$$\frac{\partial J}{\partial \theta_i} = -\frac{2}{E_x} \mathrm{Re} \sum_f \boldsymbol{e}_f^H \frac{\partial \boldsymbol{A}_f}{\partial \theta_i} \widehat{\boldsymbol{s}}_f. \tag{16}$$

For $\boldsymbol{v}_l = (\boldsymbol{p} - \boldsymbol{r}_l)/d_l$, the response derivative is

$$\nabla_{\boldsymbol{p}} a_l(f, \boldsymbol{p}) = a_l(f, \boldsymbol{p})(-d_l^{-1} - \mathrm{j}2\pi f / c_0)\boldsymbol{v}_l. \tag{17}$$

The two terms account for amplitude and delay. The joint Hessian has 2×2 blocks $H_{ij} = \partial^2 J / (\partial \boldsymbol{p}_i \partial \boldsymbol{p}_j^T)$:

$$H_{\mathrm{joint}} = \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix}. \tag{18}$$

L-BFGS-B approximates this joint curvature from position and gradient differences, without explicitly evaluating the cross-source blocks. Updating the positions together can reduce repeated conditional fits; the matched-candidate experiment in Section 4.4 measures this cost separately from candidate recovery.

Algorithm 1 MCJ-AP

Require: Observations, station positions, grid $\mathcal{G}$, candidate count M

1: Maximize Q_0 and refine the first position continuously.
2: Scan Q_{AP} conditional on this position.
3: Retain M separated local maxima in descending score order.
4: **for** each retained candidate **do**
5: Jointly minimize J from the resulting position pair, fitting all coefficients at every trial.
6: **end for**
7: Return the branch with the smallest final J.

3.2. Waveform-eliminated block NOMP

VP-BNOMP preserves residual discovery and cyclic position updates [11, 16]. During the update of source j, hold the fitted contributions of the other sources fixed:

$$\boldsymbol{y}_{j,f} = \boldsymbol{x}_f - \sum_{k \neq j} \boldsymbol{a}_f(\boldsymbol{p}_k)\widehat{s}_{k,f}. \tag{19}$$

For a trial position $\boldsymbol{p}$, eliminate the current source coefficients:

$$\widehat{q}_f(\boldsymbol{p}) = \frac{\boldsymbol{a}_f(\boldsymbol{p})^H \boldsymbol{y}_{j,f}}{\|\boldsymbol{a}_f(\boldsymbol{p})\|^2}, \tag{20}$$

$$\phi_j(\boldsymbol{p}) = E_x^{-1} \sum_f \left(\|\boldsymbol{y}_{j,f}\|^2 - \frac{|\boldsymbol{a}_f(\boldsymbol{p})^H \boldsymbol{y}_{j,f}|^2}{\|\boldsymbol{a}_f(\boldsymbol{p})\|^2} \right). \tag{21}$$

Here q_f is a complex scalar, distinct from the response vector $\boldsymbol{q}_f$. A Newton direction is computed from the gradient and Hessian of ϕ_j. Backtracking re-estimates q_f at each trial and accepts a nonincreasing local residual. The accepted contribution and explicit residual are updated together. Common least squares then re-estimates all coefficients after cyclic refinement following each source addition.

New positions are discovered by scanning the current joint residual $\boldsymbol{r}_f$ with

$$Q_{\mathrm{R}}(\boldsymbol{p}) = E_x^{-1} \sum_f \frac{|\boldsymbol{a}_f(\boldsymbol{p})^H \boldsymbol{r}_f|^2}{\|\boldsymbol{a}_f(\boldsymbol{p})\|^2}. \tag{22}$$

After common least squares, $\boldsymbol{r}_f$ is orthogonal to existing responses. The AP and residual scores then share a numerator, but use projected and unprojected response norms, respectively. Their discovery mechanisms can select different regions even though both eventually perform least-squares fitting.

Algorithm 2 VP-BNOMP

Require: Observations, grid, source count K, refinement parameters

Initialize an empty source set and the observation residual.
for $k = 1, \dots, K$ **do**
    Maximize Q_{R}, locally refine with ϕ_j, and add the source.
    repeat
        for each discovered source j **do**
            Fix the other source contributions and refine ϕ_j with safeguarded Newton updates and waveform re-estimation.
        end for
    until the cyclic energy condition or safety limit is met
    Jointly re-estimate all source coefficients and update the residual.
end for
Return the discovered positions.

3.3. Curvature and computational cost

For one local update, collect the two position coordinates in $\boldsymbol{z}$ and the real and imaginary coefficient parts in $\boldsymbol{u} \in \mathbb{R}^{2F}$. Let $\mathcal{L}_{\mathrm{loc}}(\boldsymbol{z}, \boldsymbol{u})$ denote its normalized least-squares loss and $\widehat{\boldsymbol{u}}(\boldsymbol{z})$ its coefficient minimizer. If the response is twice continuously differentiable and this minimizer is locally unique, differentiating $\nabla_u \mathcal{L}_{\mathrm{loc}} = 0$ yields

$$\frac{\partial \widehat{\boldsymbol{u}}}{\partial \boldsymbol{z}} = -H_{uu}^{-1} H_{uz}, \qquad H_{\mathrm{VP}} = H_{zz} - H_{zu} H_{uu}^{-1} H_{uz}. \tag{23}$$

All Hessian blocks are evaluated at $(\boldsymbol{z}, \widehat{\boldsymbol{u}}(\boldsymbol{z}))$. The position, coefficient and mixed blocks have dimensions 2×2, $2F\times2F$ and $2 \times 2F$, respectively. Here H_{VP} is the Hessian of the conditional single-source objective ϕ_j, with other-source contributions fixed, not of the fully concentrated joint objective $\boldsymbol{J}$. The reduced gradient equals the position gradient at the coefficient optimum [7]. Since the coefficient subproblem has positive-definite H_{uu},

$$H_{zz} - H_{\mathrm{VP}} = H_{zu} H_{uu}^{-1} H_{uz} \succeq 0. \tag{24}$$

Part of a response change caused by moving a source can be compensated by changing its waveform. Fixed-coefficient curvature retains this component; reduced curvature removes it. In positive-curvature directions with strong coupling, the former can produce small position steps.

At an interior local minimum, suppose H_{zz} and H_{VP} are positive definite and full Newton steps are used. For position error $\boldsymbol{\epsilon}$ relative to that local minimum, a fixed-curvature update of the reduced gradient has linearization

$$\boldsymbol{\epsilon}^{+} \approx (\boldsymbol{I}_2 - H_{zz}^{-1} H_{\mathrm{VP}})\boldsymbol{\epsilon}. \tag{25}$$

By (24), the iteration eigenvalues lie in $[0, 1)$ and approach one when reduced curvature is small relative to fixed-coefficient curvature. Exact reduced Newton curvature removes this linear term. Backtracking, safeguards and the outer source updates affect actual cost, which is measured separately from this local analysis.

For $\boldsymbol{u}_f = [\mathrm{Re}\, q_f, \mathrm{Im}\, q_f]^T$, the coefficient block is

$$H_{uu,f} = 2\|\boldsymbol{a}_f\|^2 \boldsymbol{I}_2 / E_x. \tag{26}$$

Independent frequency blocks allow direct accumulation of the reduced 2×2 Hessian, without a dense $2F\times2F$ matrix. To evaluate response derivatives, set $\kappa = 2\pi f / c_0$ and $\ell_l = -\log d_l - \mathrm{j}\kappa d_l$, so $a_l = \exp(\ell_l)$. Outside the 1 m protected distance,

$$\nabla \ell_l = (-d_l^{-1} - \mathrm{j}\kappa)\boldsymbol{v}_l, \tag{27}$$

$$\nabla^2 \ell_l = d_l^{-2}\boldsymbol{v}_l\boldsymbol{v}_l^T + \frac{-d_l^{-1} - \mathrm{j}\kappa}{d_l}(\boldsymbol{I}_2 - \boldsymbol{v}_l\boldsymbol{v}_l^T), \tag{28}$$

$$\nabla^2 a_l = a_l[\nabla^2 \ell_l + (\nabla \ell_l)(\nabla \ell_l)^T]. \tag{29}$$

These are complex-response derivatives with respect to real coordinates, hence the ordinary transpose. Inside the protected distance, that station's derivatives are zero; its boundary is nonsmooth. Analytic derivatives were checked against finite differences, automatic differentiation and small-matrix calculations.

Table 1 lists leading costs. Here G is grid size, T counts continuous objective/gradient calls, T_M sums them across branches, I counts source cycles, T_{loc} counts local calls, and I_B counts SBL iterations. For segmented covariance methods, S and F_s denote segments and retained segment-frequency bins. Both curvature rules have linear cost in F; their measured savings arise from step and cycle counts.

Table 1

Leading operation counts; joint coefficient re-estimation is included in complete runtime.

Operation	Cost
Common DFT	$\mathcal{O}(LN \log N)$
MUSIC/MVDR	Statistics $\mathcal{O}(F_s S L^2 + F_s L^3)$; scan $\mathcal{O}(F_s G L^2)$
Joint coefficient fit	$\mathcal{O}(F(LK^2 + K^3))$
MCJ-AP	$\mathcal{O}(FGLK + T_M F(LK^2 + K^3))$
VP-BNOMP	Scans $\mathcal{O}(KFGL)$; local updates $\mathcal{O}(IKT_{\mathrm{loc}} FL)$
Multistart/GA	$\mathcal{O}(TF(LK^2 + K^3))$
Dense wideband SBL	$\mathcal{O}(I_B(FGL^2 + FL^3))$

For fixed source count, accepted MCJ-AP steps and VP-BNOMP block updates do not increase their respective fitting residuals; common coefficient re-estimation also cannot increase the residual. The resulting nonnegative objective sequences converge. This property concerns objective values, not global position convergence. AP checks a projected joint gradient, whereas VP-BNOMP checks cyclic energy change. The mechanism experiments retain the corresponding matched stopping rules.

4. Simulation results

4.1. Design, baselines and evaluation

The stations are located at $(500, 0)$, $(0, 500)$, $(-500, 0)$ and $(0, -500)$ m. Their maximum baseline is $R = 1,000$ m and the search region is $[-1,000, 1,000]^2$ m^2. Table 2 lists the common acquisition and search settings. Source positions are continuous. The two independent BPSK waveforms have independent symbol sequences, initial phases and timing, and the same center frequency and bandwidth.

Table 2

Common acquisition and search settings.

Quantity	Setting
Stations and sources	Four synchronized single-antenna stations; known $K = 2$
Propagation	Relative amplitude $1/d$ and baseband delay; $c_0 = 299792458$ m/s
Signal	BPSK; RRC shaping, 10 MHz symbols, roll-off 0.25; 12.5 MHz bandwidth
Acquisition	50 MHz, 8,192 samples, 163.84 μs
Processing band	$[-7, 7]$ MHz; 2,293 full-record DFT bins
Power	Weak-source mean in-band SNR 6 dB; $\Delta = 0, 3, 6, 9, 12, 15$ dB
Search	Common 20 m grid, 101×101 points; continuous refinement
Candidates	3×3 local maxima, minimum separation 30 m; $M = 1$–5

Within each source configuration and realization, all six power gaps reuse the same waveforms and noise. Only source A is scaled. The band is fixed in advance, includes 0.75 MHz guards around the nominal signal band, and is shared by every method; no per-source spectrum or per-station RMS normalization is supplied.

The five-candidate setting was fixed in the evaluation protocol before the 2,400-observation run. It was retained unchanged for the new source configurations. Candidate-count comparisons evaluate this fixed choice's cost and recovery, rather than establish an optimal value of M.

Twenty source configurations cover both sources inside the station convex hull, A outside/B inside, A inside/B outside, and both outside, with five configurations per category (Fig. 1). Generation excludes sources within 150 m of a station, 100 m of the center or 50 m of a station-pair line, and requires at least 300 m between sources. Positions are not selected by localization performance. All methods use two preassigned realizations per configuration and all power gaps, giving 240 observations. Core methods use 20 realizations, giving 2,400 observations including those 240. Frozen methods are then tested at 16 new source configurations with five realizations and six gaps, giving 480 observations. Receiver coordinates remain fixed; Appendix A supplies new source coordinates and seeds.

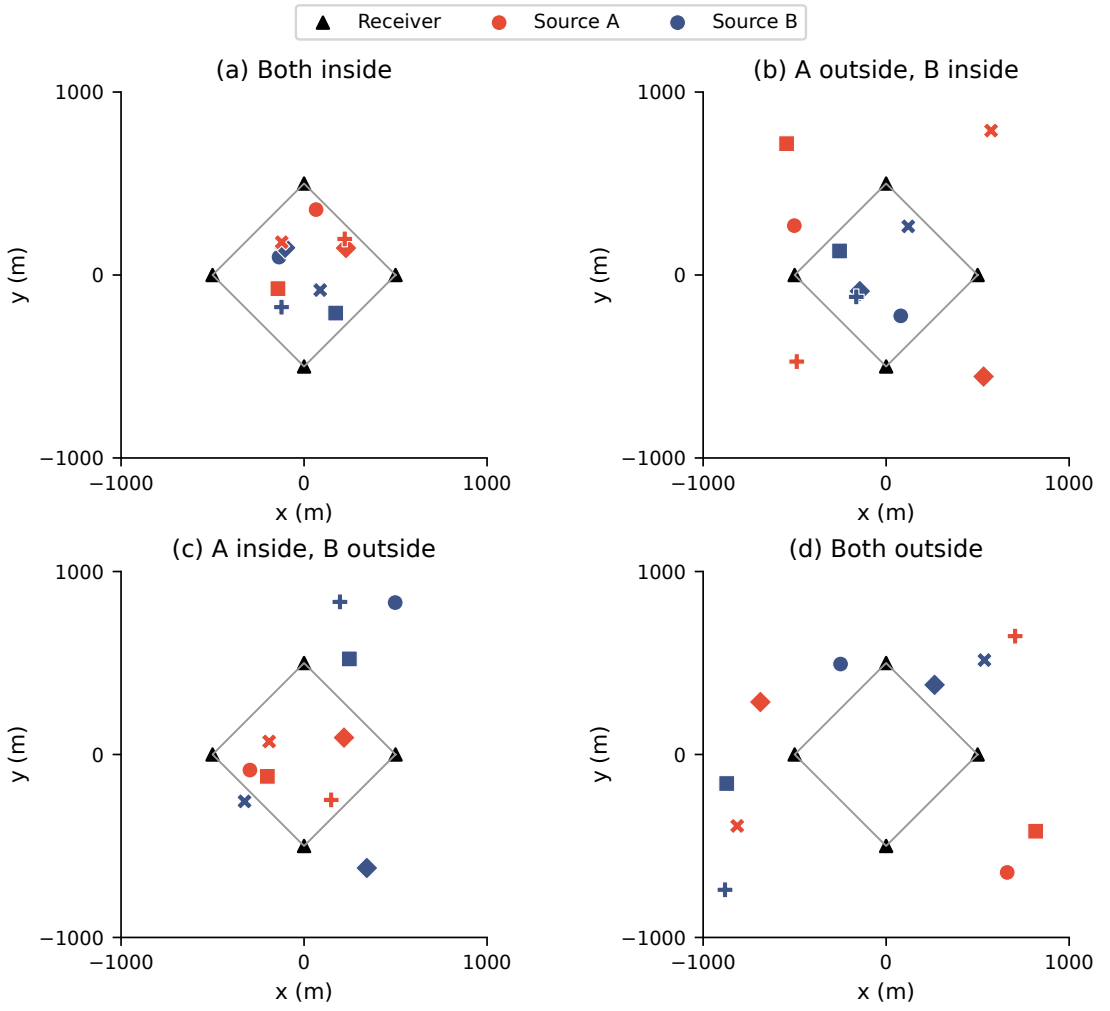


Figure 1: Twenty source configurations, five per category. Black triangles denote stations. Matching shapes pair source A (red) and B (blue) within each panel; the diamond is the station convex hull.

All methods share IQ data, propagation information, double-precision calculations and the initial grid. Their adapted objectives are specified below; native initialization and continuous refinement are retained.

AP branch refinements share projected-gradient tolerance 10^{-8} on the normalized objective, no objective-difference stopping term, and a 100-iteration local limit. VP-BNOMP and its fixed-coefficient reference use cyclic energy tolerance 10^{-8}, at most five Newton steps per local call, a 50 m step cap, a 0.01 m local small-step exit and a 10,000-cycle safety limit. The different gradient and energy conditions are not equivalent optimization accuracies. Complete-method comparisons use their frozen native rules; joint-refinement and curvature comparisons match the relevant stopping conditions explicitly.

For n observations indexed by i, one squared-distance assignment per observation [10] determines errors for all reported position metrics:

$$\pi^* = \arg\min_{\pi} \sum_{k} \|\widehat{\boldsymbol{p}}_{\pi(k)} - \boldsymbol{p}_k\|^2, \quad e_{ik} = \|\widehat{\boldsymbol{p}}_{i,\pi^*(k)} - \boldsymbol{p}_{ik}\|. \tag{30}$$

RMSE is $\sqrt{n^{-1}\sum_i e_{ik}^2}$ and the empirical CDF is $n^{-1}\sum_i \mathbf{1}(e_{ik} \le t)$. Both include all observations. Two-source success requires $\max_k e_{ik} \le 20$ m, a prescribed tolerance of $0.02R$ distinct from grid spacing. Invalid outputs remain failures in the denominator; all reported batches returned two finite positions.

Complete runtime includes the DFT, band selection, transfer, search, coefficient estimation, refinement and necessary output checks, with GPU synchronization. Adapter trajectory recording and final diagnostics are included where present; input-file reading, disk writes and offline statistics are excluded. Warm-up is separate. Runtime comparisons use the same batch for speedup ratios. The environment is Windows, Python 3.10.20, NumPy 2.2.6, PyTorch 2.8.0+cu129, an RTX 4070 Ti SUPER and 20 CPU threads, with double-precision responses and objectives. Resampling retains the six gaps of one realization as a group. Candidate-cost and curvature intervals condition on fixed layouts; broader hierarchical intervals are identified separately below.

4.2. Baseline objectives and implementation

LS peaks continuously refines two separated maxima of (11), without jointly fitting two sources. For covariance methods, split the same record into S nonoverlapping segments of $N_s = N/S$ samples. With segment DFT $\boldsymbol{z}_{bf} \in \mathbb{C}^L$, normalized by $1/N_s$,

$$\widehat{\boldsymbol{C}}_f = S^{-1} \sum_{b=1}^{S} \boldsymbol{z}_{bf} \boldsymbol{z}_{bf}^H. \tag{31}$$

The zero-mean model makes correlation and covariance equal. Segment noise variance is σ_t^2/N_s; only the same physical processing band is retained. No snapshots are pooled across realizations.

MUSIC requires $K < L$, independent response columns, full-rank source covariance and spatially white noise [15]. Let $\boldsymbol{E}_{n,f}$ contain the $L-K$ smallest-eigenvalue eigenvectors. The wideband score and noise-subtracted weights are

$$S_{\mathrm{MU}}(\boldsymbol{p}) = \left[\sum_f w_f \boldsymbol{q}_f^H \boldsymbol{E}_{n,f} \boldsymbol{E}_{n,f}^H \boldsymbol{q}_f\right]^{-1}, \tag{32}$$

$$\widetilde{w}_f = \max\{\mathrm{tr}(\widehat{\boldsymbol{C}}_f) - L\sigma_t^2/N_s, 0\}, \qquad w_f = \widetilde{w}_f / \sum_v \widetilde{w}_v. \tag{33}$$

Weights are uniform if their sum vanishes. Known K fixes the subspace dimension, but finite-sample accuracy depends on effective snapshots and signal power at each bin.

MVDR minimizes $\boldsymbol{h}_f^H \boldsymbol{C}_{f,\delta} \boldsymbol{h}_f$ subject to $\boldsymbol{h}_f^H \boldsymbol{q}_f = 1$, where $\boldsymbol{h}_f \in \mathbb{C}^L$ is a combining vector [2]. Loading and wideband averaging give

$$\begin{aligned} \boldsymbol{C}_{f,\delta} &= \widehat{\boldsymbol{C}}_f + \delta \, \mathrm{tr}(\widehat{\boldsymbol{C}}_f) \boldsymbol{I}_L / L, \\ S_{\mathrm{MV}}(\boldsymbol{p}) &= F_s^{-1} \sum_f [\boldsymbol{q}_f^H \boldsymbol{C}_{f,\delta}^{-1} \boldsymbol{q}_f]^{-1}. \end{aligned} \tag{34}$$

The dimensionless loading δ scales mean diagonal power. MVDR-known uses known relative responses and this frequency average, distinguished from the original DPD-HR formula [14].

For SBL, define the dictionary $\boldsymbol{D}_f = [\boldsymbol{q}_f(\boldsymbol{g}_1), \ldots, \boldsymbol{q}_f(\boldsymbol{g}_G)]$ and frequency coefficients $\boldsymbol{z}_f \sim \mathcal{CN}(0, \boldsymbol{\Gamma})$, independent across frequencies with shared nonnegative powers $\boldsymbol{\Gamma} = \mathrm{diag}(\boldsymbol{\gamma})$ [12]. The covariance, posterior mean, negative log evidence and fixed-point update are

$$\boldsymbol{C}_f = \sigma_f^2 \boldsymbol{I}_L + \boldsymbol{D}_f \boldsymbol{\Gamma} \boldsymbol{D}_f^H, \qquad \boldsymbol{\mu}_f = \boldsymbol{\Gamma} \boldsymbol{D}_f^H \boldsymbol{C}_f^{-1} \boldsymbol{x}_f, \tag{35}$$

$$\mathcal{E}(\boldsymbol{\gamma}) = F^{-1} \sum_f [\log\det \boldsymbol{C}_f + \boldsymbol{x}_f^H \boldsymbol{C}_f^{-1} \boldsymbol{x}_f], \tag{36}$$

$$\gamma_j^+ = \gamma_j \left[\frac{\sum_f |\boldsymbol{d}_{fj}^H \boldsymbol{C}_f^{-1} \boldsymbol{x}_f|^2}{\sum_f \boldsymbol{d}_{fj}^H \boldsymbol{C}_f^{-1} \boldsymbol{d}_{fj}} \right]^{1/2}, \tag{37}$$

where $\boldsymbol{d}_{fj}$ is column j of $\boldsymbol{D}_f$. Evidence backtracking controls updates. Every five iterations, eight active positions are optimized off-grid, allowing 20 m displacement per coordinate; L-BFGS-B uses coordinate scale 1,000 m and at most 20 iterations. Final peaks use individual γ_j. Every five iterations, stopping requires normalized evidence change below 10^{-5}, relative ℓ_1 power change below 1%, and output/active-position motion below 0.05 m on two consecutive checks; the cap is 300 iterations.

Full AP repeatedly searches the conditional objective and refines positions, using five candidates and up to six outer cycles. Multistart ML minimizes $\boldsymbol{J}$ from four unique spectral peak combinations and four Sobol spatial starts, removing unordered duplicates. GA-ML encodes four coordinates, with 128 individuals, four elites, size-three tournament selection, simulated binary crossover (probability 0.9, index 15), and polynomial mutation (per-coordinate probability 0.25, index 20) [6]. It stops after 30 stagnant generations or 200 generations, then jointly refines the best individual.

The fixed-coefficient BNOMP reference (FC–FC) differentiates

$$L_{\mathrm{FC}}(\boldsymbol{p}) = E_x^{-1} \sum_f \| \boldsymbol{y}_f - \boldsymbol{a}_f(\boldsymbol{p}) q_f^{(0)} \|^2, \tag{38}$$

where $q_f^{(0)}$ is fitted at the current position, held fixed for differentiation and reference backtracking, and re-estimated after acceptance. The multi-frequency block structure follows [16]; Hessian repair, bounds and backtracking are numerical safeguards. The two tags specify curvature and trial acceptance, respectively. VP-BNOMP uses reduced curvature and reduced acceptance (VP–VP). Since original NOMP combines fixed-gain curvature with re-estimated-gain acceptance [11], we also test fixed-coefficient curvature with reduced acceptance (FC–VP) to isolate the curvature change.

MUSIC uses four segments; MVDR-known uses eight and $\delta = 0.01$. AP local calls allow 25 line-search trials and 100 iterations; alternating refinement allows 12 sweeps and initial single-source refinement uses relative tolerance 10^{-12}. BNOMP floors Hessian eigenvalues at 10^{-9} and tries step factors $1, 1/2, 1/4, 1/8, 1/16$ within the search bounds. Its other limits and AP's common gradient tolerance were specified above. Parameters are fixed before the reported comparisons.

4.3. Common-data comparison and expanded evaluation

Table 3 compares all methods on the same 240 observations. MCJ-AP matches the complete recovery of full AP and SBL at substantially lower complete cost. SBL has a wider weak-source error distribution. The two BNOMP versions recover the same 228 observations, with lower mean cost for the waveform-eliminated version.

Table 3

Common 240 observations. RMSE in m; runtime mean, median and P95 in s. Every metric includes all observations.

Method	Success/240	RMSE A	RMSE B	Mean	Median	P95
LS peaks	38	191.982	845.184	0.2368	0.2181	0.3297
WB-MUSIC	151	93.515	646.892	0.2015	0.1893	0.2719
MVDR-known	75	320.684	804.082	0.1925	0.1807	0.2625
Full AP	240	0.876	1.596	8.9076	9.1483	12.2348
BNOMP (FC–FC)	228	0.906	89.706	2.9416	1.4625	8.1989
VP-BNOMP	228	0.875	86.177	0.5907	0.3921	1.3971
Multistart ML	45	256.061	776.935	1.2227	1.0868	2.5595
GA-ML	185	279.056	501.231	8.0702	8.1102	9.0229
WB-SBL	240	1.096	3.743	44.1296	36.8237	97.7075
MCJ-AP ($M = 5$)	240	0.876	1.596	0.7081	0.6347	1.0232

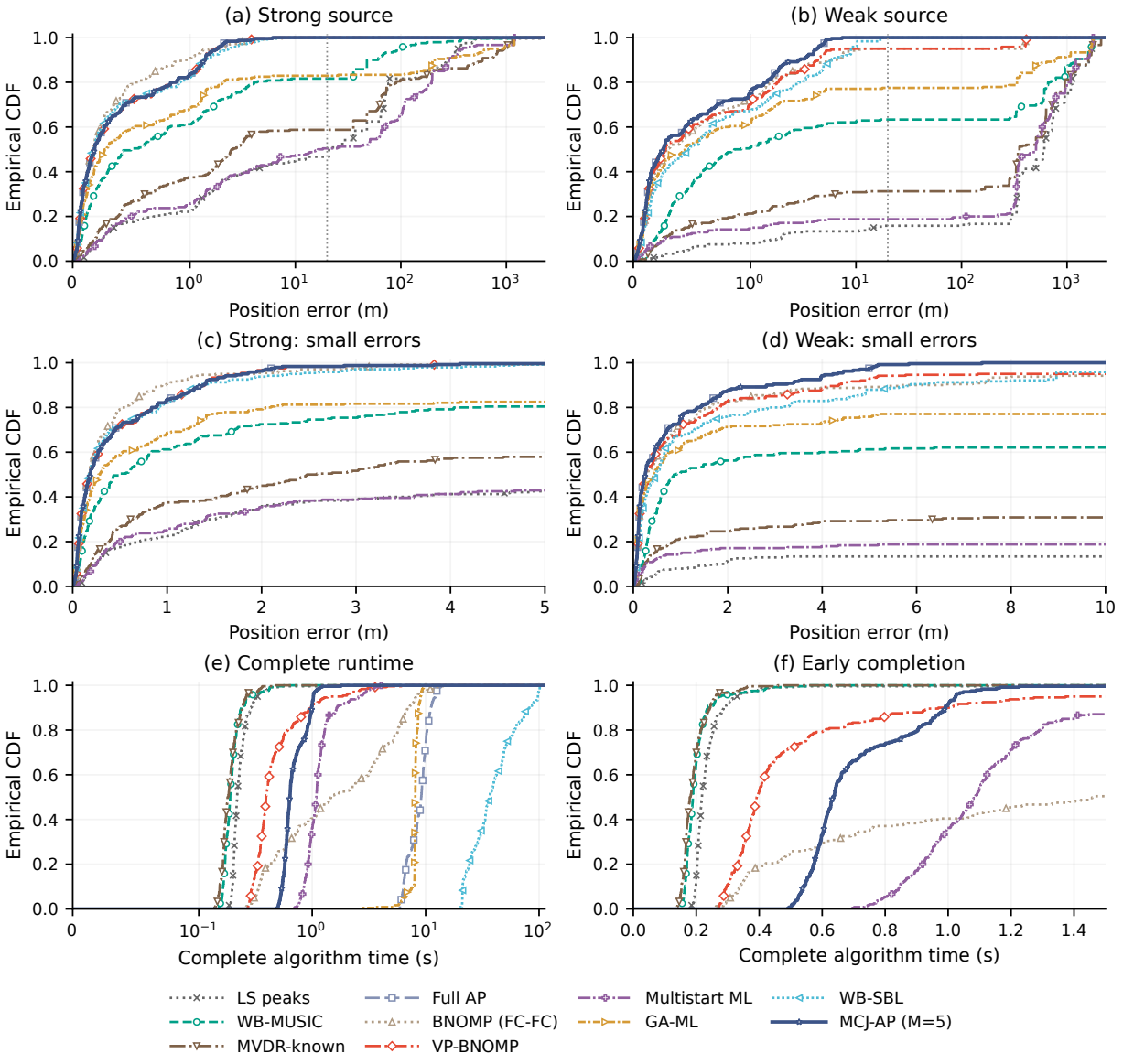


Figure 2: Error and complete-runtime CDFs on the common 240 observations. The error panels show full ranges and small-error enlargements; the time panels show the full range and early completion. Full-range axes are linear near zero and logarithmic beyond their thresholds. Vertical error lines indicate 20 m. All curves retain the full denominator.

Figure 2 distinguishes typical precision from large-error tails. The AP error curves nearly coincide. VP-BNOMP is accurate on most observations, but retains a distant weak-source tail and occasional long runtimes. Spectral methods are fast, yet their failures involve large errors rather than small threshold crossings.

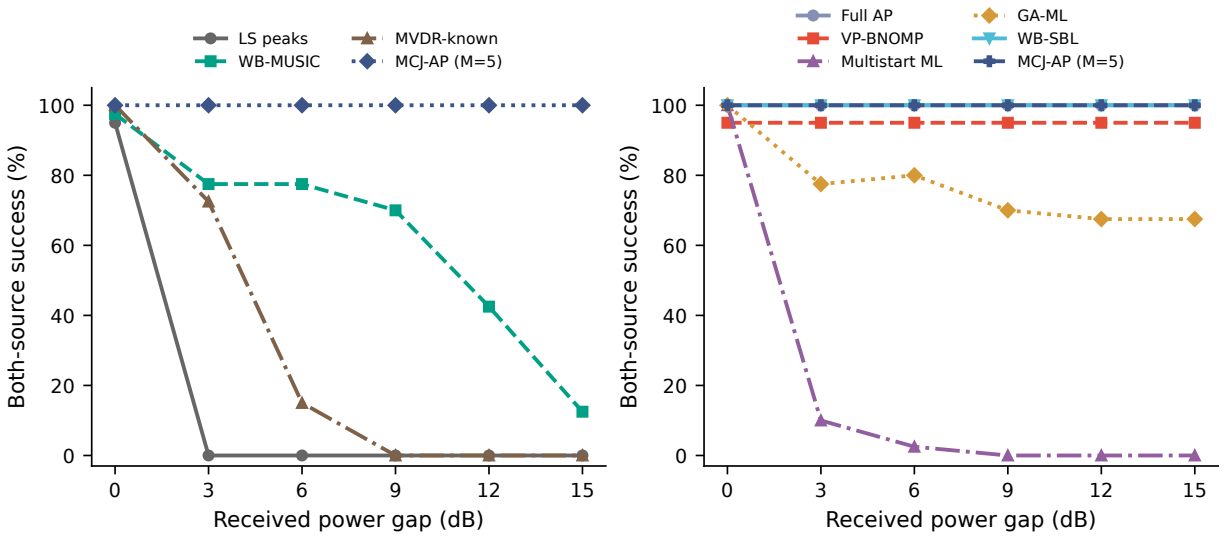


Figure 3: Power-gap sweep on the common subset, 40 observations per gap. Both panels use the same five-candidate AP reference.

As the power gap increases, MUSIC and MVDR-known lose recovery (Fig. 3); Section 4.7 examines whether finite-window mismatch explains the tested failures. Full AP and multistart ML optimize the same joint objective but have very different outcomes, demonstrating the importance of initialization and search. GA reaches its generation limit in 208 observations and SBL in 14; the other methods have no limit exits. All outputs remain included.

Saved outputs on the common subset first show complete success at 0.75, 1.5 and 30 s for five-candidate AP, full AP and SBL, respectively. These checkpoints describe recorded results, not executable early-stopping rules. VP records only its final complete output in this batch, so the different recording schedules do not support a strict anytime ranking.

Table 4

Expanded 2,400-observation evaluation. RMSE in m and complete time in s. VP-BNOMP timing is from a separate same-input batch; controlled speedups are evaluated in paired batches.

Method	Success/2400	RMSE A	RMSE B	Mean	Median	P95
AP ($M = 1$)	2280	0.897	107.265	0.3958	0.3700	0.5104
AP ($M = 2$)	2280	0.897	103.698	0.4704	0.4329	0.6340
MCJ-AP ($M = 5$)	2376	0.896	46.984	0.7180	0.6495	1.0267
VP-BNOMP	2278	0.892	74.193	0.5855	0.3693	1.2470

On 2,400 observations, five candidates increase AP success from 95% to 99% (Table 4). Hierarchical resampling gives a 95% interval of [96.83%,100%] for its success rate. The four-percentage-point increase over two candidates has interval [0,12.17] percentage points, reflecting concentration in one configuration. The common 240 subset and expanded cohort are reported separately rather than counted as independent observations.

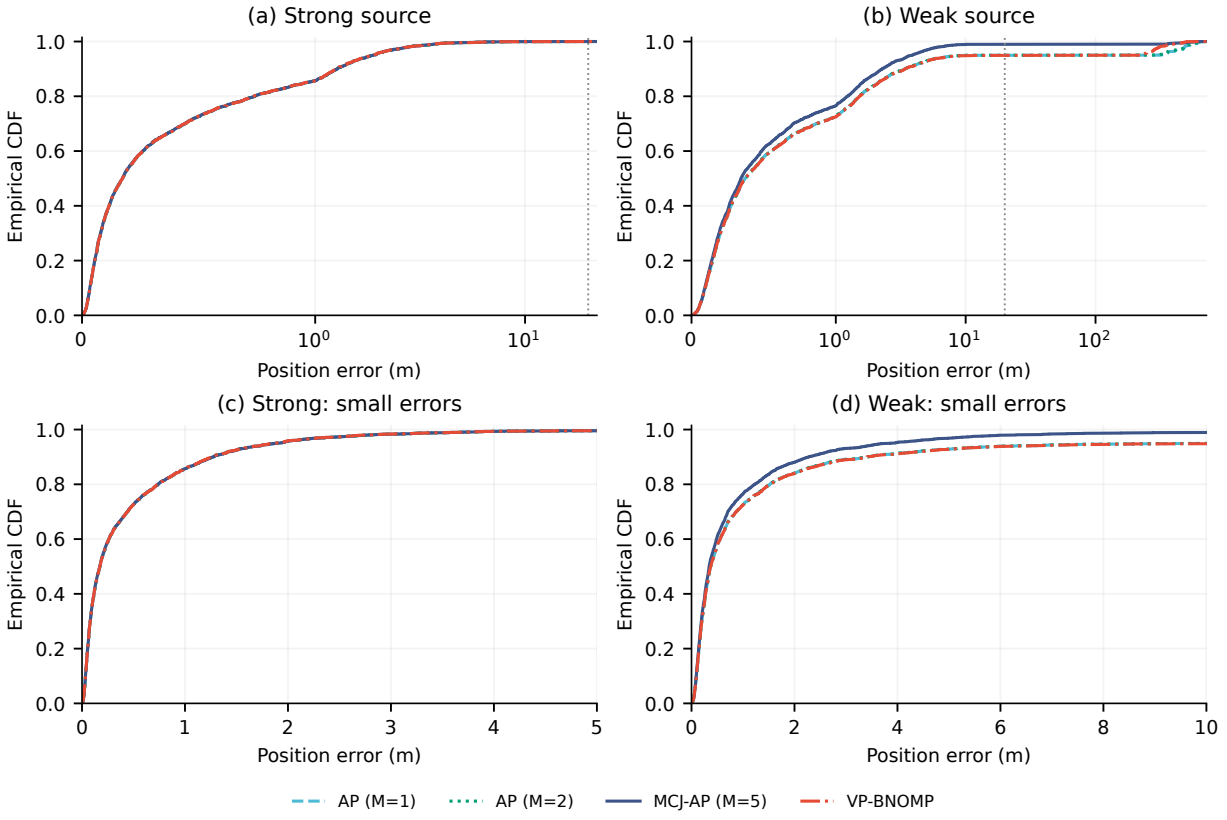


Figure 4: Strong- and weak-source error CDFs for all 2,400 core observations. Upper panels cover full ranges and lower panels enlarge small errors. All observations, including failures, remain in each curve.

The strong-source CDFs nearly coincide (Fig. 4). Small weak-source errors are also similar, whereas tails differ. Five-candidate AP has weak-source P95 of 3.85 m but RMSE of 46.98 m. Configuration 4 accounts for all 24 failures and over 99.7% of its weak-source squared error in each power bin.

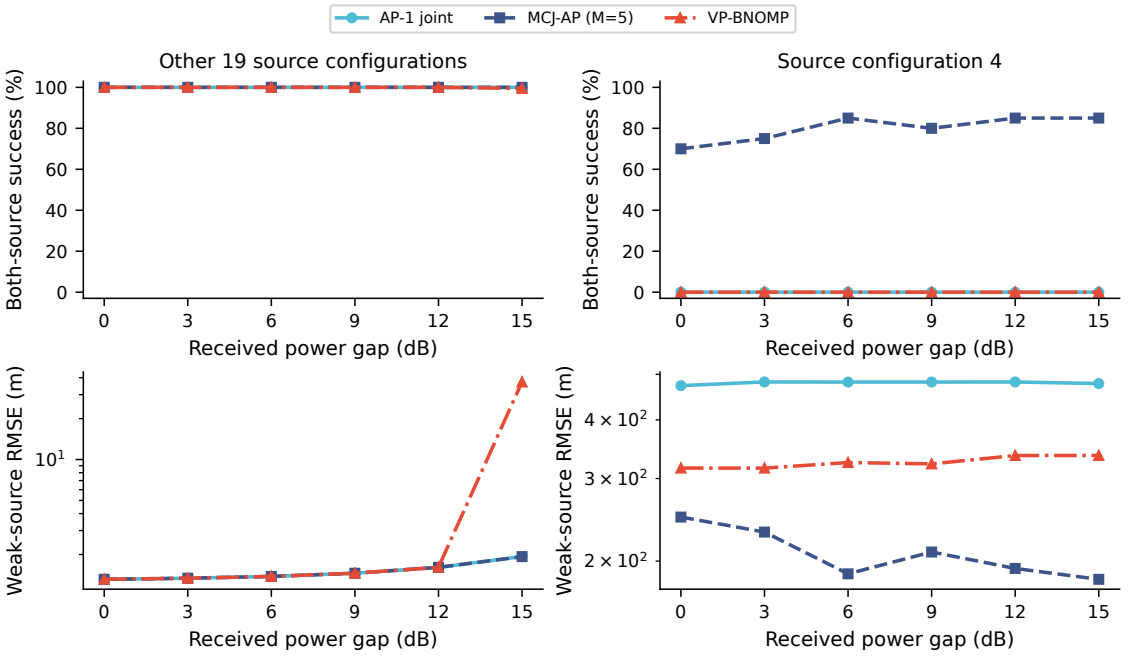


Figure 5: Power-gap results for the other 19 configurations (2,280 observations) and configuration 4 (120 observations). This diagnostic grouping leaves all observations in the overall results.

In the other configurations, AP weak-source RMSE rises from 1.31 to 1.93 m across the power sweep (Fig. 5). VP has two failures in configuration 1 at 15 dB, producing 15.25 m RMSE over those 19 configurations versus about 1.52 m for AP. In configuration 4, VP reduces weak-source RMSE from single-candidate AP's 479.66 m to 325.07 m without meeting the 20 m tolerance. Configuration 4 drives the nonmonotone overall RMSE; geometry is unchanged across power bins.

4.4. Candidate coverage and joint-refinement cost

Lower-ranked conditional candidates account for all 96 additional AP recoveries. Table 5 records the first successful refined branch in the original conditional-score order. Success uses the true positions only for offline evaluation; MCJ-AP selects solely by the final residual J. The recoveries all occur in configuration 4 and involve 18 of its 20 waveform/noise realizations. Its other 24 observations have no successful branch among the five refined candidates.

Table 5

First successful refined branch by initial conditional-score rank, all 2,400 observations. Success requires both matched errors to be at most 20 m.

First successful rank	1	2	3	4	5	None
Observations	2280	0	1	8	87	24

Of the 96 recoveries, 87 require the fifth-ranked candidate. Final residual selection chooses a successful branch in all 2,376 observations that contain one, with no loss of success elsewhere. Thus lower initial scores do not preclude effective continuous solutions in this configuration.

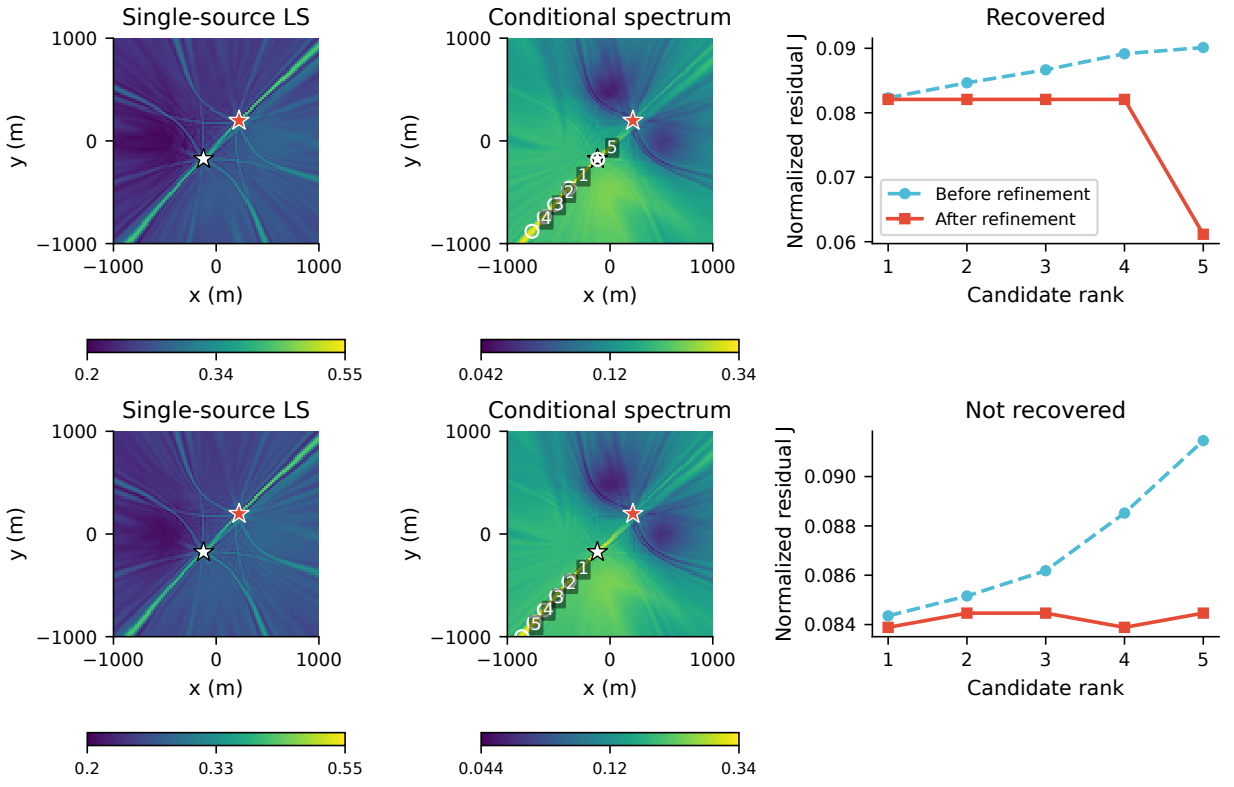


Figure 6: Two post hoc cases from configuration 4 at 0 dB. Columns show the single-source spectrum, conditional spectrum and pre-/post-refinement branch residuals. The upper case is recovered by the fifth candidate; the lower case has no effective candidate. Lines in the right panels connect branch indices, not optimization trajectories.

In the upper case of Fig. 6, the first four branches enter a distant region; the fifth gives a lower final residual and approximately 0.09 m weak-source error. In the lower case, even the nearest candidate is about 397 m from source B and refinement does not recover it. These cases illustrate recovery through a lower-ranked branch and failure of all five local fits, respectively.

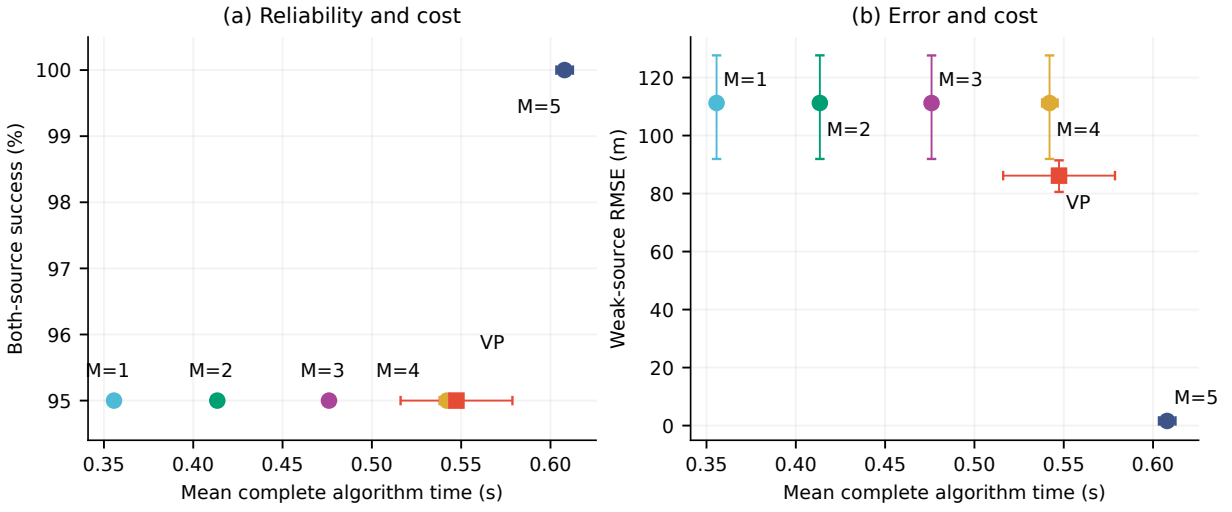


Figure 7: Independent complete solves for AP with $M = 1$–5 and VP-BNOMP on the same 240 observations, timed in one interleaved batch. Error bars are 95% paired intervals resampling realizations within fixed source configurations while retaining the six gaps together.

In this batch, $M = 1$–4 each succeeds on 228 observations; mean runtime increases from 0.3556 to 0.5420 s and weak-source RMSE remains about 111.2 m. VP-BNOMP has 228 successes, 86.18 m weak-source RMSE and 0.5474 s mean runtime. Five-candidate AP succeeds throughout at 0.6079 s and 1.60 m weak-source RMSE. VP median time is 0.3674 s versus single-candidate AP's 0.3514 s; its P95 is 1.3147 s versus five-candidate AP's 0.7201 s. Among 228 common successes, VP is faster than five-candidate AP in 197; five candidates recover the other 12 observations. These results distinguish typical speed from recovery and runtime tails. Saved full-cohort prefixes give 2,281 and 2,289 successes for three and four candidates, respectively; these are position-only replays.

Table 6

Paired alternating and joint refinement of the same five candidates on 2,400 observations. Both selected outputs meet the same projected-gradient tolerance.

Refinement	Success	RMSE B/m	Mean time/s	J calls	Gradient calls
Alternating	2376	46.983	1.0840	204.21	198.21
Joint	2376	46.984	0.7441	120.36	114.36

Joint refinement reduces the cost of processing the same five candidates by 31.36% (Table 6). Both methods use the same full projection objective and selected-output projected-gradient tolerance; success/failure identities agree observation by observation. Strong-source RMSE is about 0.896 m and weak-source P95 about 3.85 m for both.

Gradient calls fall by 42.30%, consistent with fewer optimization calls rather than a looser stopping rule. The paired hierarchical 95% interval for joint-minus-alternating time is $[-0.3691, -0.3121]$ s. Every configuration has lower mean cost, with savings from 11.42% to 48.04%. Two unselected alternating branches reach the local safety limit and remain included. These savings concern refinement at fixed candidate count; the added candidates' recovery and extra cost are measured separately in Fig. 7.

4.5. Isolating waveform-eliminated curvature

The common-data fixed-coefficient/VP comparison in Table 3 changes both local curvature and the trial acceptance objective, reducing mean runtime by 79.92%. To isolate curvature, a separate paired batch applies the same waveform-eliminated objective ϕ_j to trial acceptance. FC–VP computes directions with H_{zz} and VP–VP with H_{VP}. Both use identical source-discovery and initialization procedures, conditional coefficient and gradient formulas, safeguards, reduced-objective backtracking and cyclic energy stopping criteria. Only the local curvature construction is changed; the resulting positions and residual paths can differ.

Table 7

Curvature comparison with a common waveform-eliminated acceptance objective, 240 paired observations. Update and cycle counts are means per observation.

Version	Success	RMSE A/m	RMSE B/m	Mean time/s	Updates	Cycles
FC–VP	228/240	0.906	89.706	5.574	1498.82	810.90
VP–VP	228/240	0.875	86.177	0.905	135.55	47.97

Success/failure identities agree and all outputs stop under the native energy rule. Mean complete runtime falls by 83.77%, with fixed-layout paired 95% interval [81.73%,85.58%]. Median time falls from 2.836 to 0.567 s and P95 from 16.161 to 2.538 s. VP–VP is faster in 170 of 240 observations: the mean saving is driven especially by reductions in slow cases. The associated update and cycle reductions support the curvature mechanism in (25).

Mean core times are 5.5483 and 0.8987 s; recording and final checks add 0.0254 and 0.0059 s. Audits of 480 historical/current VP pairs find matching inputs, positions, objectives, counts, stopping reasons and saved paths despite different absolute wall times. Each speedup therefore uses its own paired batch: 79.92% measures the complete local-update change, while 83.77% isolates curvature under common acceptance. The latter interval uses 4,000 within-configuration paired bootstrap draws (seed 20261020), retaining all six gaps of a realization. Equal energy thresholds do not impose a common final joint-gradient tolerance; Section 4.7 examines that distinction.

Terminal quality is evaluated using the same fully concentrated $\boldsymbol{J}$ and the infinity norm of its box-projected gradient in meter coordinates. At an absolute objective-comparison tolerance of 10^{-10}, VP–VP gives a lower $\boldsymbol{J}$ in 199 pairs

and an approximately equal value in 41, with none higher; at 10^{-8} the counts are 167, 73 and zero. Absolute objective differences have median 3.34×10^{-7} and P95 7.37×10^{-6}. The largest difference, 1.97×10^{-3}, occurs in an observation where both methods fail localization. Projected-gradient norms are smaller for VP–VP in all 240 pairs, with medians 1.30×10^{-7} versus 2.03×10^{-6} for FC–VP. Thus the measured cost saving is accompanied by no deterioration in these terminal fitting and stationarity measures under the native stopping rule; localization quality remains assessed by the position errors.

4.6. Frozen methods at new source positions

Table 8 and Fig. 8 assess the 16 new source configurations. Both AP variants recover all 480 observations with nearly identical errors, while five candidates cost approximately 1.96 times as much as one. The first branch is already effective, so additional branches do not increase success. Together with configuration 4, this shows the fixed candidate budget's tradeoff: it preserves recovery opportunities when the leading candidate is ineffective, but does not reduce work when the first branch already suffices.

Table 8

New source configurations, 480 observations; receiver positions unchanged. RMSE in m and complete runtime in s.

Method	Success/480	RMSE A	RMSE B	Mean	Median	P95
AP ($M = 1$)	480	0.958	2.194	0.461	0.456	0.545
MCJ-AP ($M = 5$)	480	0.958	2.194	0.906	0.893	1.078
VP-BNOMP	390	202.882	206.261	1.051	0.629	4.245

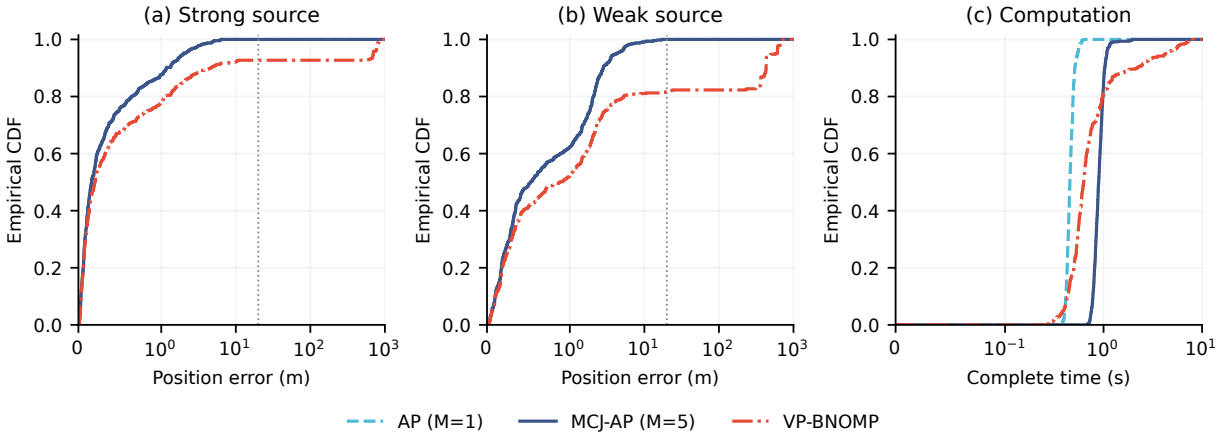


Figure 8: Error and complete-runtime CDFs on the 480 new observations. AP error curves nearly overlap. All observations are retained; error reference lines mark 20 m. Full-range axes are linear near zero and logarithmic beyond the stated thresholds.

VP-BNOMP has 90 failures in new configurations 8, 9, 14 and 15: 30, 25, 30 and 5, respectively. Every gap has 15 failures; 55 involve only B, 33 both sources and two only A. All five realizations are represented. Failed observations average 1.448 s versus 0.959 s for successful ones; they stop by energy stability, with larger final objectives than AP. They already fail at the first saved full pair and never subsequently succeed. Thus neither a discarded saved success nor safety-limit truncation explains the failures. Because recording begins after one two-source cycle, it does not distinguish failed discovery from a particular update.

4.7. Finite-window and numerical checks

Eight development observations (four configurations at 0 and 15 dB) provide saved clean source components. At the true positions, the finite-window residual and expected projected noise energy are

$$E_{\mathrm{win}} = \sum_{b,f} \|\mathbf{\Pi}^{\perp}_{\boldsymbol{A}_f} \boldsymbol{z}^{(0)}_{bf}\|^2, \qquad E_n = SF_s(L-K)\sigma_t^2/N_s. \tag{39}$$

Every $\boldsymbol{A}_f$ is full column rank. For one, four and eight segments, respectively, the residual fractions are 0.209–0.382%, 0.756–1.387% and 1.529–2.854%; the corresponding E_{win}/E_n ranges are 0.035–0.882, 0.126–3.210 and 0.238–6.611. Shorter windows increase the mismatch.

Rectangular and periodic Hann windows were compared with a 4,096-sample Hann window using four overlapping frames. Windowed noise variance is $\sigma_t^2 \sum_n w[n]^2/N_s^2$; overlapping frames are not independent snapshots. At 15 dB, the long window reduces E_{win}/E_n to 0.076–0.294, but MUSIC and MVDR-known still recover only 1/4 and 0/4

observations, retaining the original failures. Both recover 4/4 at 0 dB. Projecting each clean source onto its true response before restoring the original noise removes almost all model residual, yet none of the seven failed method–observation pairs recovers. This truth-assisted diagnostic is excluded from ranking. Finite-window mismatch alone does not explain the tested failures.

Numerical convergence was also checked separately from localization error. Continuing native VP paths on eight development observations until the joint projected gradient reaches 10^{-8} keeps success at 8/8, changes weak-source RMSE from 3.22 to 3.25 m and raises mean complete time, including gradient checks, from 0.382 to 0.860 s.

The FC–FC small-step mechanism was examined on eight development observations, separately from the formal FC–VP/VP–VP pair. With cyclic energy tolerances 10^{-8}, 10^{-7} and 10^{-6}, successes are 8/8, 7/8 and 6/8; mean times are 5.478, 2.318 and 0.971 s, and derivative calls are 2679.5, 1133.9 and 467.4. Removing only the small-step local exit gives 8/8 at 8.011 s and 4758.4 calls; waveform elimination at 10^{-8} gives 8/8 at 0.436 s and 70.4 calls. All 21,436 reference updates are accepted, 83.67% shorter than 0.01 m, without Hessian repair, step clipping, backtracking reduction or boundary clipping. At two outside-source diagnostic positions, raw steps increase from 0.0138/0.0111 m to 11.25/32.26 m after elimination, before backtracking. At four saved positions, analytic reduced Hessians match automatic differentiation within about 1.4×10^{-18} maximum absolute difference. These checks motivate the formal curvature pair; they are not substitutes for its complete cost measurements.

5. Conclusions

This paper addresses candidate selection and local update cost in direct localization of two unknown overlapping wideband sources. MCJ-AP retains conditional peaks until both positions have been jointly refined. The saved branches show why initial score alone can discard effective solutions: most additional recoveries in the tested difficult configuration require the fifth-ranked peak. Joint L-BFGS-B reduces the cost of refining the same candidates under a common projected-gradient stopping tolerance. VP-BNOMP incorporates waveform elimination into the local position curvature, and the common-acceptance comparison confirms lower average cost and fewer updates without worsening the measured terminal residual and projected-gradient norm. The new source configurations separate these benefits from complete-flow reliability: extra AP candidates add no recovery there, and VP retains failures. Under the stated synchronized, known-response baseband model, the evidence supports delayed branch selection for candidate recovery and reduced curvature for local computational efficiency. Future work can examine candidate-budget allocation and unknown source count.

A. New source coordinates

Table 9 lists the new positions in meters to six decimal places. Categories 1–4 correspond to both inside, A outside/B inside, A inside/B outside, and both outside the receiver convex hull. Geometry and waveform seeds are 20261020071 and 20261020072; realization identifiers are 1000–1004. Configuration numbers are one-based. These positions obey the same geometric exclusions as the original cohort and were not screened by method performance.

Table 9

New source positions. A is power-scaled; B and the noise are fixed.

No.	Category	x_A	y_A	x_B	y_B
1	1	-209.742960	-211.672998	-135.426339	121.578497
2	1	-115.361316	76.809910	67.769590	-192.728293
3	1	-208.559801	115.499630	268.616751	-92.726639
4	1	173.459164	-110.441904	-168.154517	-145.923194
5	2	-926.529606	771.380729	-198.481435	152.456653
6	2	-628.314566	-929.843027	-55.519139	-297.939120
7	2	-195.757938	-496.419920	157.528284	224.885259
8	2	-734.894082	-306.033731	-249.535479	71.485648
9	3	-64.564529	351.783227	663.100329	428.023655
10	3	-166.451414	260.105867	823.995321	431.176047
11	3	74.808116	299.580428	-702.622192	474.464614
12	3	-162.472651	-112.907043	765.537745	-808.531366
13	4	313.792247	602.900501	700.222844	-274.728424
14	4	482.010461	239.752856	888.180085	90.857032
15	4	-374.838719	599.269525	240.800462	663.866438
16	4	624.598019	542.011213	-163.287363	904.989973

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